

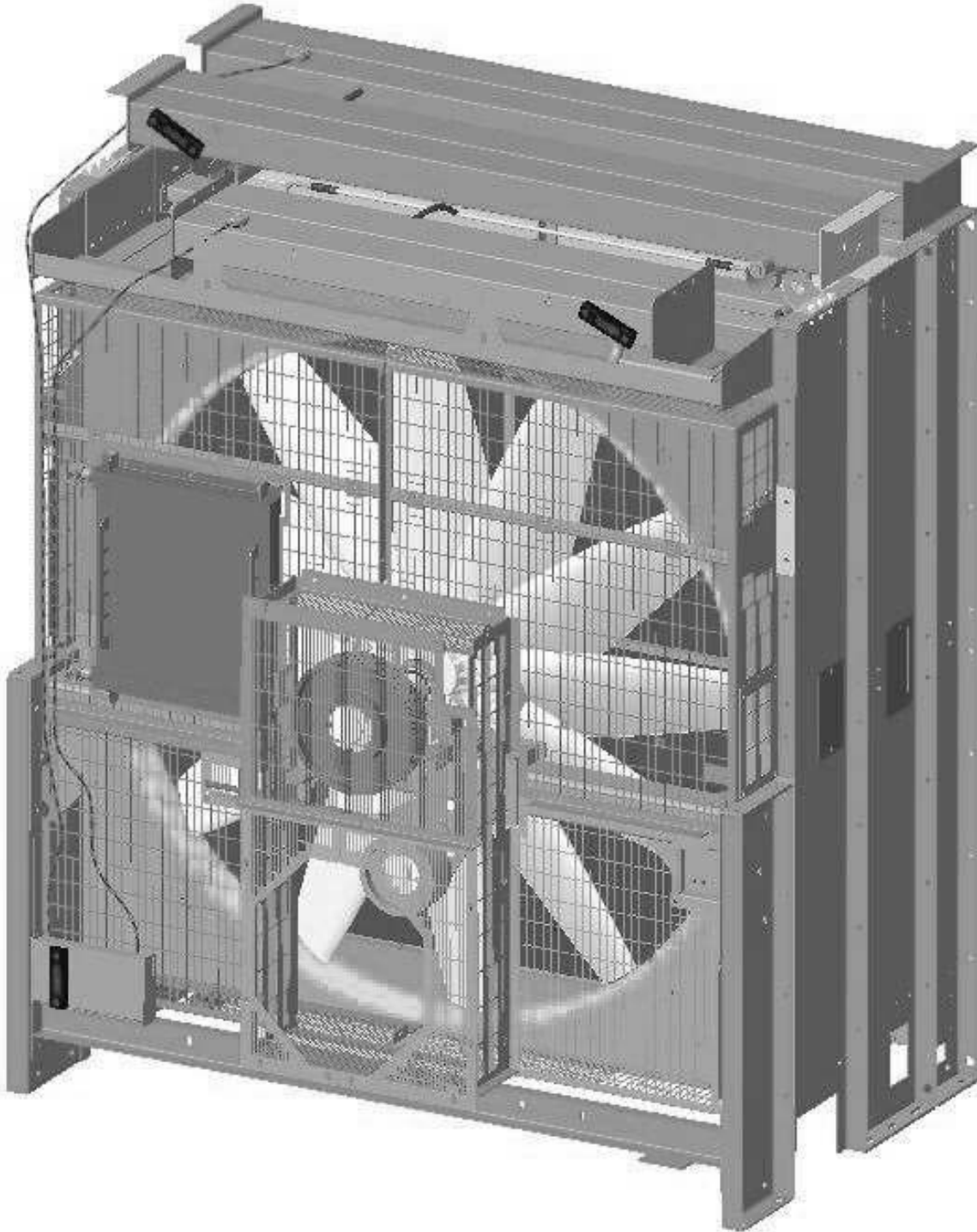


**GENERIC INSTALLATION,
SERVICE, MAINTENANCE AND
REPAIR MANUAL
For Belt Drive Radiators**

PI-10-25-00-36

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Issue/Mod: **0 17742**



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1. Introduction.

1.1. General description.

This radiator consists of one or two fluid circuits and a mechanically driven cooling fan coupled to the output pulley of the engine (Fig.2).

The radiator should not be used for anything other than the application described above. The radiator is specifically designed for this engine and application and as such may not perform as expected on other engines resulting in potential damage or loss of performance.

1.2. About this document.

Safe and efficient operation of the radiator can only be achieved if the equipment is properly operated and maintained. Perceived poor performance or failure of the radiator can often be caused by a failure to follow fundamental rules and precautions.

The purpose of these instructions is to provide the user with information specific to the use and operation of the radiator.

It is important to read all instructions fully before proceeding with any Service, Maintenance or Repair tasks. If in doubt please contact the manufacturer, see Section 1.3.

Note that original radiator installation to the Genset is covered in the Genset manufacturer's assembly literature.

The information contained within this document is based upon information available at the time of issue, however the manufacturers policy of continuous product improvement may mean that the product could change at any time. The user should make sure before commencing any work that they have the latest information available, please contact the manufacturer if in doubt, see Section 1.3

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1.3. Safety.

It is the end installer's responsibility to ensure that only competent persons are employed to carry out any tasks on the radiator.

Important safety points are:

- Isolate Genset electrically before attempting any tasks. Ensure battery pack is disconnected.
- Before lifting components check the area and route and use 2 persons to lift heavy or large components when lifting aids are unavailable.
- Working at heights. Radiator parts should not be used to stand on. Use harnesses when working on top of radiator.
- Be aware of potentially sharp edges on some steel components.
- Radiator pipework and cooling surfaces are very hot during operation. Care should always be taken when the machine is running. Ensure radiator is cooled before starting work.
- Beware coolant and/or fuel oil contact. Risk of skin irritation if hands and/or clothing become contaminated.
- Genset environments will be noisy when running and will be above 85dBA. Wear suitable hearing protection.
- Beware of trapping fingers, long hair and clothing in rotatable parts of the Genset even when it is not running i.e. fan drive belts and pulleys when turning the fan by hand.
- Operating pressure of the Genset is detailed on the Wabtec customer drawing.

1.4. Guarding

Some parts of the guarding have been designed to allow engine components to pass through or fit inside the openings. The clearances designed for these openings conform to BS EN ISO 14120:2015 & BS EN ISO 13857:2008 when the radiator is connected to the engine.

It is the installers responsibility to ensure the appropriate guarding standards are applied as Bearward Engineering have no control over the design of the Genset and any modifications the may be made to it. If this radiator is fitted to any engine that isn't covered within this instruction, then it is the installers responsibility to ensure the guarding conforms to the abovementioned standards.

The removable guarding has been designed with hey-hole slots to allow the controlled removal of the guard. When the fixings all loosened the guard with drop to the bottom of the slot and then can be removed.

When refitting the guard, ensure it is lifted to the top of the slots before tightening the fixings. Once all guarding is refitted, check all gaps and edges to ensure it is fitted correctly and not sitting at the bottom of the slots.

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DO NOT	   	Do not grease or clean equipment while it is moving. Do not eat or drink while in vicinity of equipment. Do not switch on while working on equipment. Do not remove guarding while equipment is moving.
SAFETY	      	Risk of electrocution. Risk of slippery surfaces. Risk of loud noise. Risk of burns from hot surfaces. Risk of trapping fingers or loose clothing. Potential headroom hazard. Risk of contact with toxic substances i.e. coolant and fuel oil.
MANDATORY	       	Wear correct PPE. Use correct lifting techniques. Wash hands thoroughly after work.

Fig.1. Safe operating practice risk notifications



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1.5. Contact details.

Bearward Engineering, A Division of Wabtec UK LTD
Bearward Engineering
Main Rd
Far Cotton
Northampton
England
NN4 8HJ

Tel: +44 (0)1604 762851

General Web Enquiries: www.Wabtec.com

For Sales Enquiries: BEN_Sales@Wabtec.com

For Service Enquiries: BEN_Service@Wabtec.com

For Spare parts Enquiries: BEN_Spares@Wabtec.com

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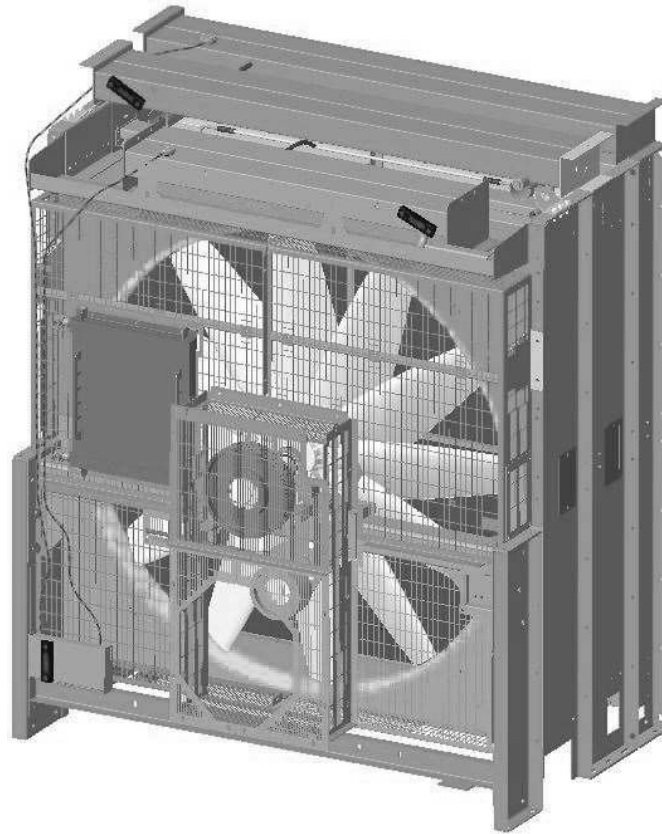


Fig.2. Front view of a Belt Drive radiator showing general arrangement.

2. Installation.

This section should be read in conjunction with the installation drawings provided as part of the documentation pack and the Genset manufacturers installation literature.

2.1. General installation

2.1.1 The radiator will be shipped standing vertically or horizontally and secured to a transport pallet. Remove any ancillary components that are secured to the radiator for transport purposes and carefully put to one side.

2.1.2 Check any loose accessories supplied against customer drawings.

2.1.3 Ensure appropriate lifting equipment is available to lift and position the radiator in place. Unit information including mass and overall dimensions can be found in Chapter 5. Specifications of this document and the accompanying installation drawings.

2.1.4 After removing the fixings that secure the radiator to the pallet, attach lifting hooks to the angled brackets circled in Fig.2. The radiator can be placed into position.

Coarse positioning should be carried out according to Genset manufacturers guidelines with careful attention being paid to ground level and flatness. Over large distances, a small angle of misalignment can result in large positional errors at the extremes of the product. This usually manifests in pipe connection gaps being too large or clashing and / or drive system alignment difficulties.

2.1.5 Accompanying instruction PI-00-10-00-02 provides additional generic information regarding labelling and lifting.

2.1.6 Follow genset manufacturer's instructions regarding vibration reduction. However, it is recommended that anti-vibration mountings are used to reduce vibration at the radiator.

2.1.7 Once in position all connections can be made as per the installation drawings:

Hose clips are to be fitted according to: PI-15-00-00-02
Torque applied as per: BPS-20-35-10-02

2.1.8 Ensure that the unit is adequately earth bonded in accordance with EN60204-1:2006 +A1 2009, clause 8.2.1.



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2.1.8 Pulley and belt alignment.

Follow PI-10-25-00-30 for the pulley alignment and belt tensioning.
For this work the recommended tools are: Laser + Contitech tensioner.

Frequencies for belt tension are detailed on each customer drawing as a Hz value.

Pulley alignment using TKBA40 laser alignment tool: PI-10-25-00-31
Belt tension using Contitech tension gauge: PI-10-25-00-30

2.2. Filling:

Fill radiator using the coolant recommended by the engine manufacturer and percentage mix recommended by Bearward for Glycol and coolant.

DO NOT OVERFILL this radiator. Only fill to the “max coolant level” labels on both circuits.

Run engine to clear any potential airlocks. Shut down and allow to cool before checking the level gauges and topping up if required.

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3. Service requirements.

Application	No. of runs/year	Run hours/year	Recommended coolant check interval	Recommended brg grease interval (hrs)	Recommended core clean interval	Expected radiator life (years)
Commercial standby	52	200	each use	250	As required *	20
Commercial prime (limited)	200	750	each use	250	As required *	20
Commercial prime (unlimited)	360	3000	each use	250	As required *	5
Commercial continuous	360	8280	each use	250	As required *	4

* Core clean interval is dependent upon visible degree of fouling of the core. This will very much depend upon local environment and Genset usage. As a general rule it is best to ensure a visible core fin edge profile and also unobstructed view through the core matrix. However the maximum effective period between core cleaning cycles will be determined by radiator cooling performance and its effect upon Genset engine overheating.

Fig.3. Table listing service intervals.

3.1. Checking coolant condition and level.

Important note. Ensure coolant temperature is at ambient room temperature before checking level; this check will usually be done prior to starting the Genset. This will avoid problems associated with apparent high coolant levels due to coolant expansion and will also avoid the risks of hot coolant scalds. Never check coolant level when engine is running.



Fig.4. Coolant level inspection sight glasses